Question 1: Hypergeometric and Binomial Distributions

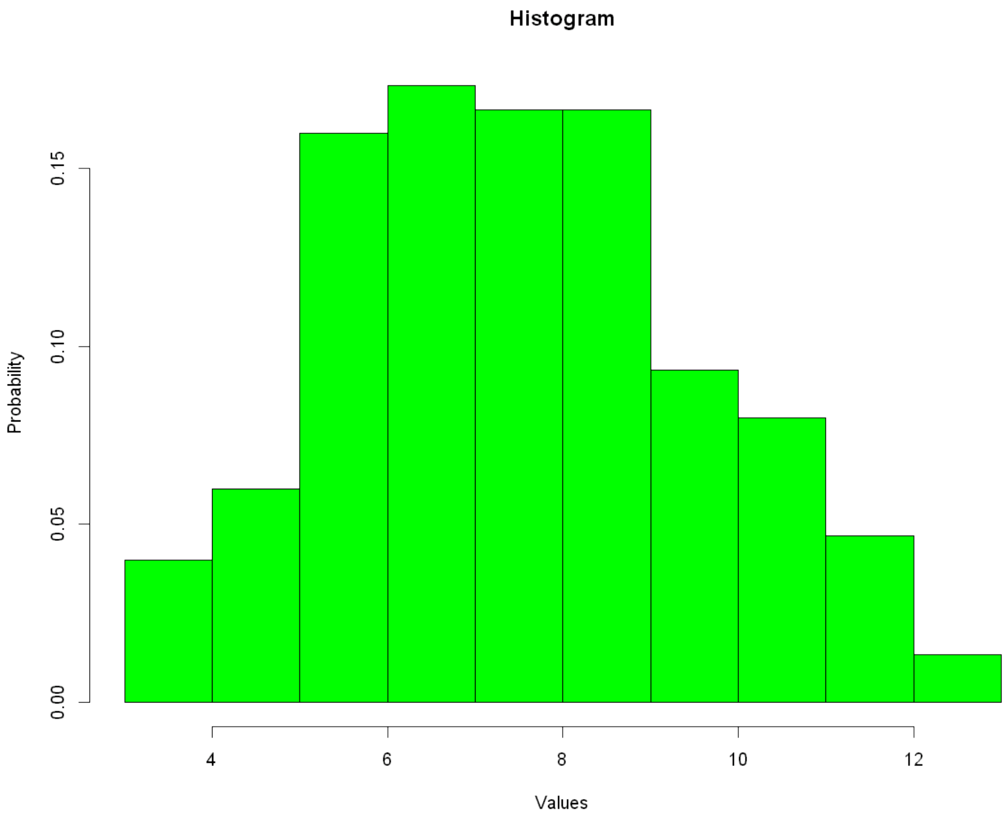
Part 1: Hypergeometric Distribution

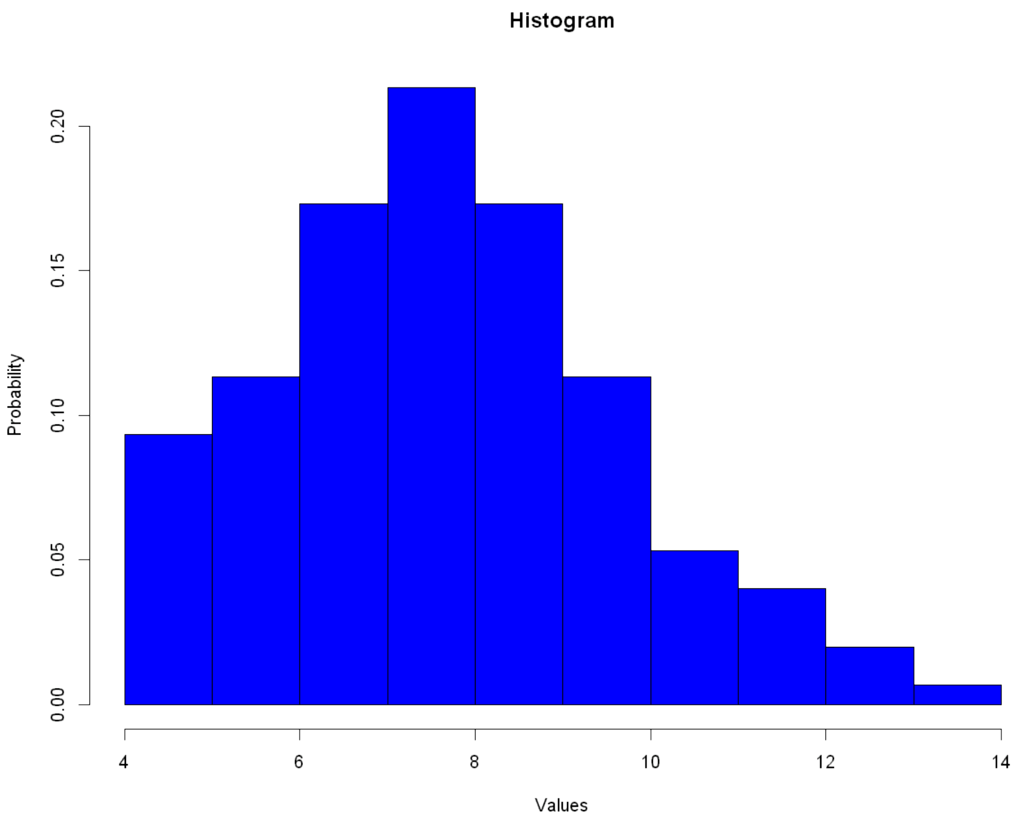
We simulate the distribution by doing the experiment 150 times. First I implement this distribution manually using for loop.

Or we can use the predefined function instead. There is the rhyper() function in r which does exactly the same.

Now I plot both the results using a histogram.

They are exactly the same, but generated with two different methods. Because the distributions are random in essence they are probabely distributed differently.



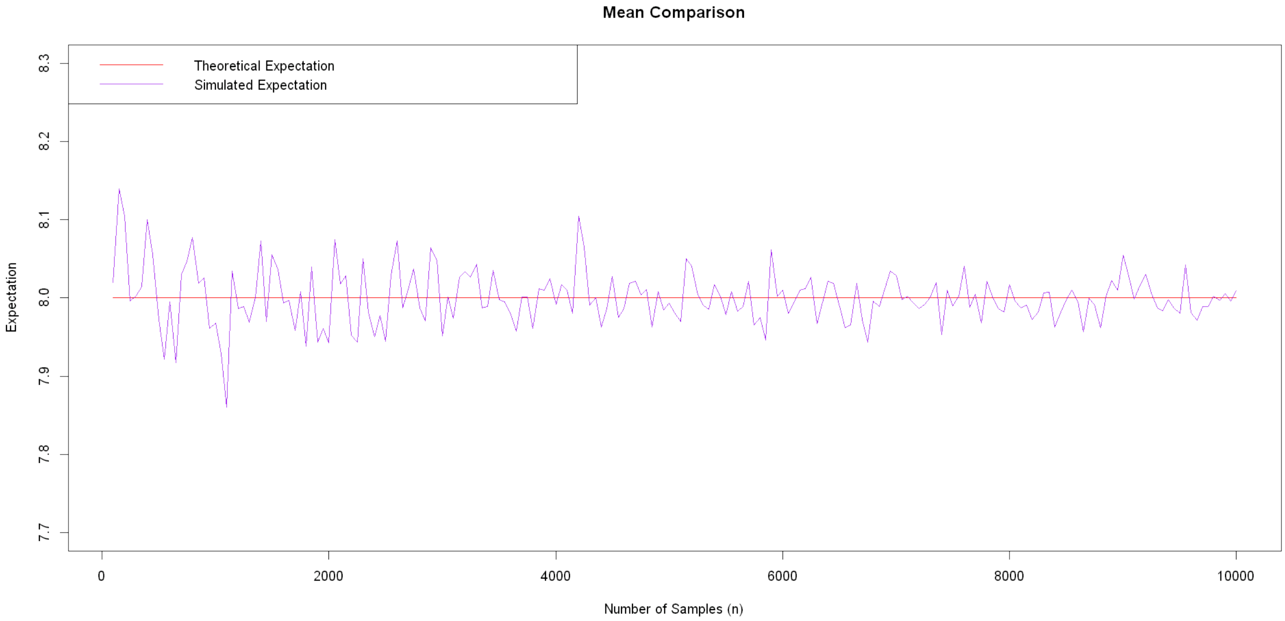


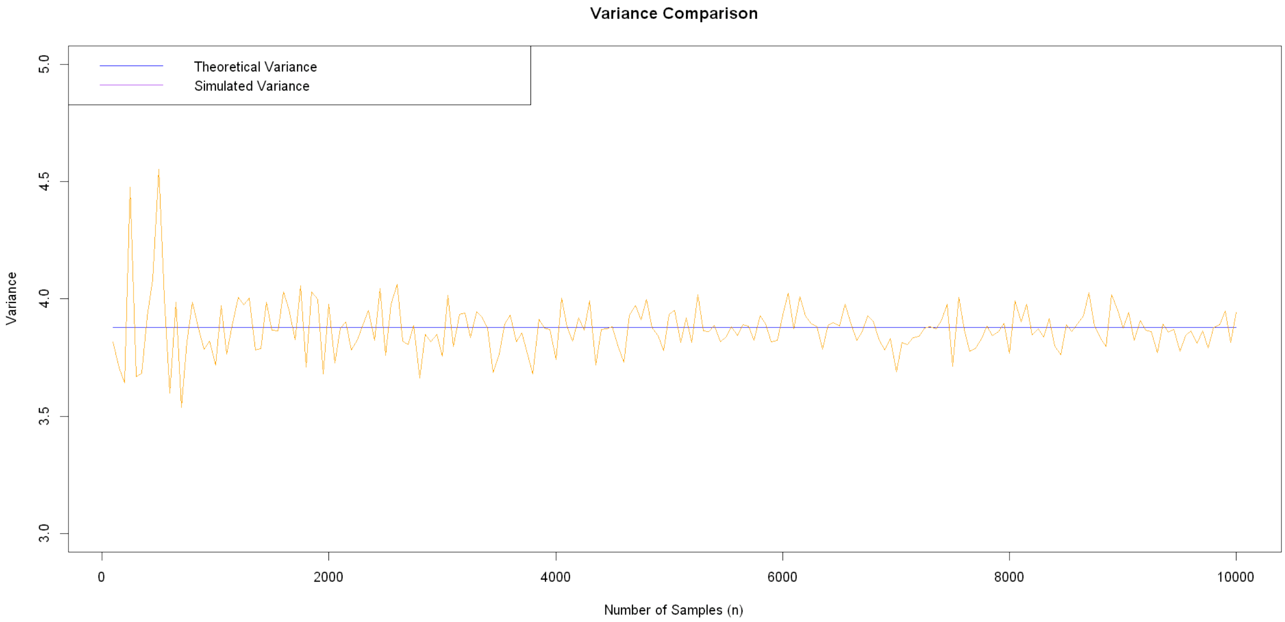
Part 2: Theoretical and Simulated Variances and Expectations

This function returns a R data frame which has both theoretical and simulated values for variances and expectations.

Part 3: Plotting Part 2

In this part I'm gonna plot the simulated and theoretical values to convert them.



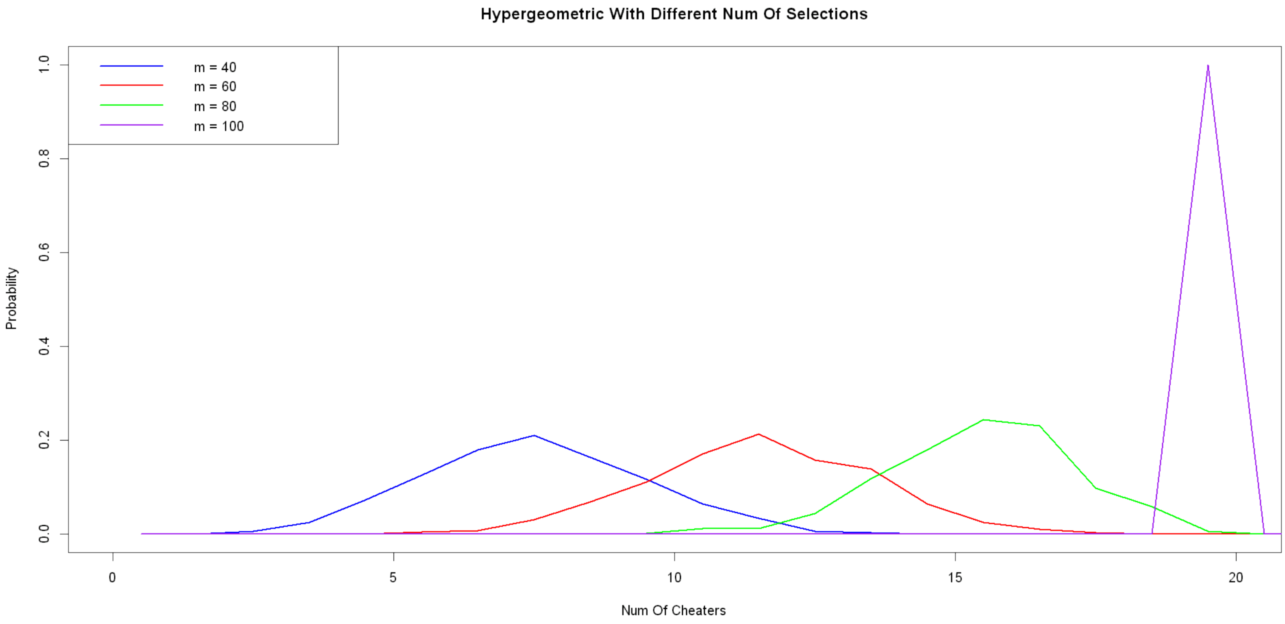


To conclude we can say that if we increase the n(make it close enough to infinity), then the values for mean and variance aquired by simulations are going to be converge to their theoretical values.

Part 4: Keeping n Constant and Changing m

I'll keep the variable n equal to 1000 and change the variable num_of_selections over [40, 60, 80, 100].

```
Warning message in hist.default(temp_distribution, breaks = seq(0, N + 1, by = 1), :
"argument 'probability' is not made use of"
Warning message in hist.default(temp_distribution, breaks = seq(0, N + 1, by = 1), :
"argument 'probability' is not made use of"
Warning message in hist.default(temp_distribution, breaks = seq(0, N + 1, by = 1), :
"argument 'probability' is not made use of"
Warning message in hist.default(temp_distribution, breaks = seq(0, N + 1, by = 1), :
"argument 'probability' is not made use of"
```



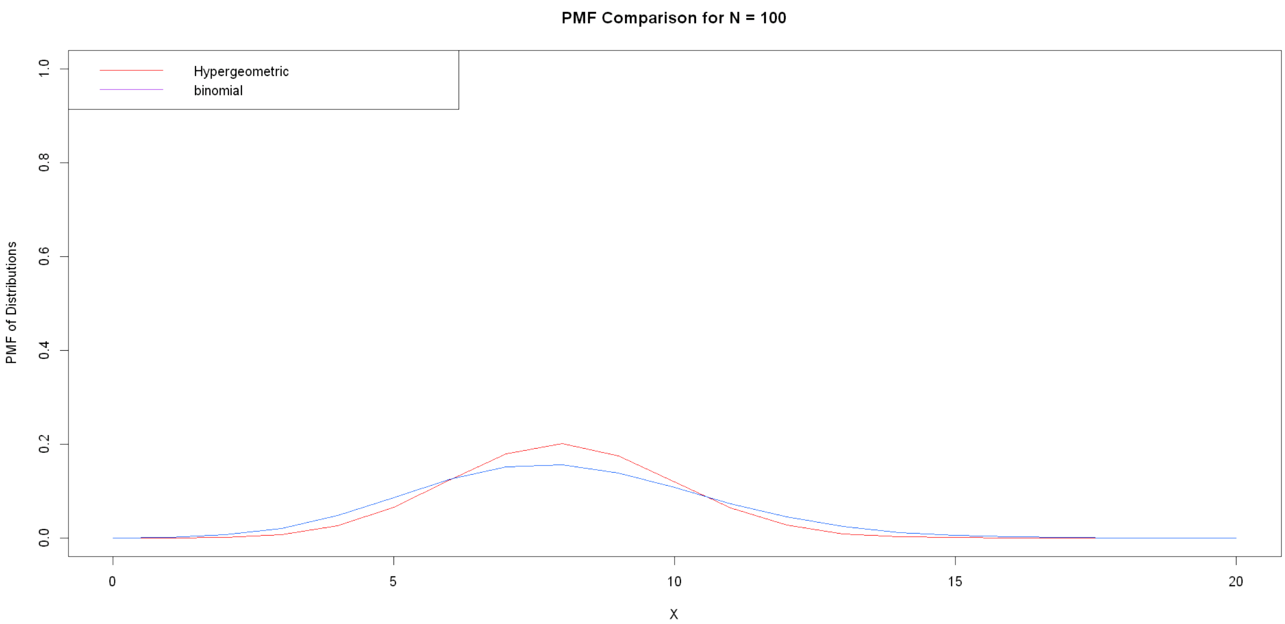
As we can see, when m equals 100 because we have selected all of the districts the number of cheaters we have encountered is equal to 20 which is equal to the whole number of cheaters; becuase we have checked all of the districts.

For each of other values for m there is a peak approximately on m * N/K which is the expectation of distribution.

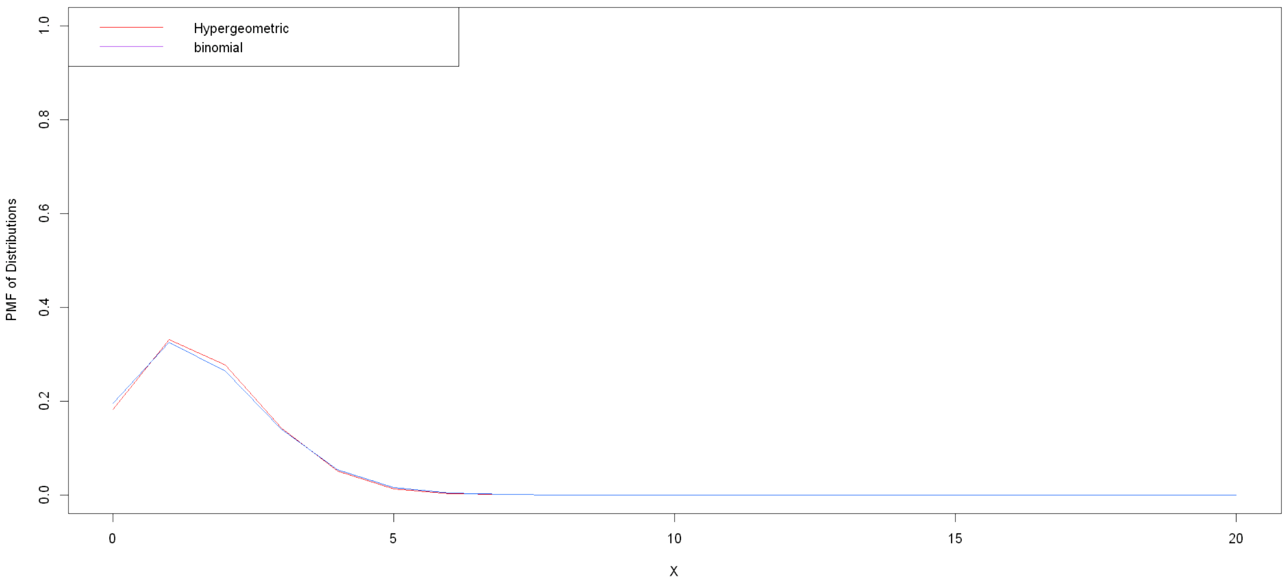
Part 5: Functions For Calculating PMF For Hypergeometric and Binomial Distributions

Part 6: Hypergeometric and Binomial PMFs

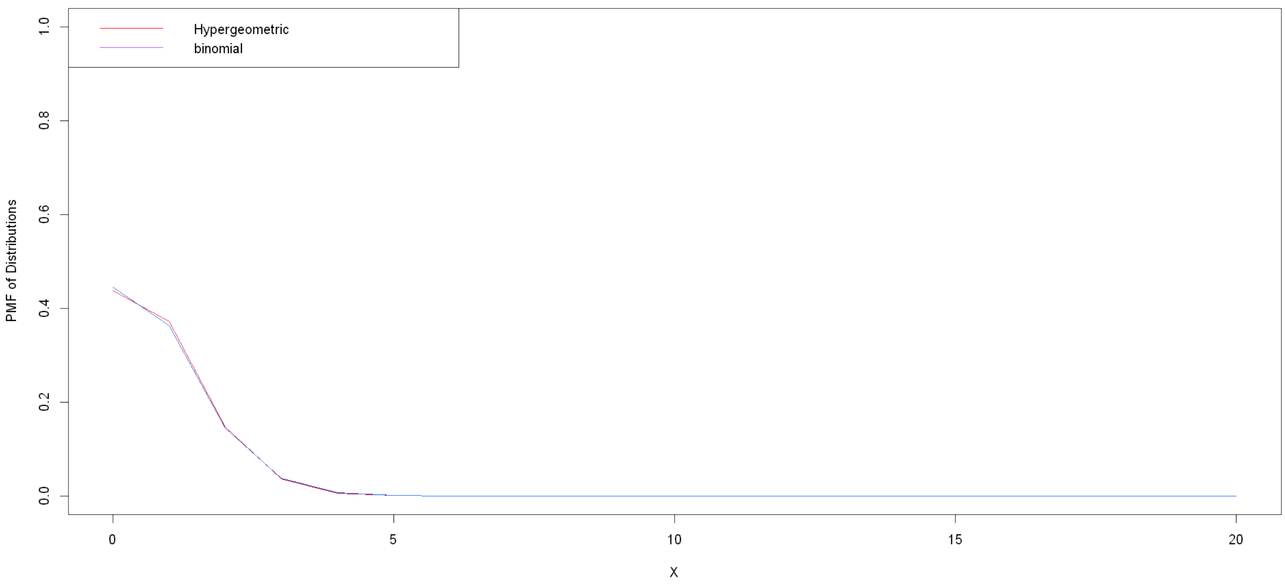
In this part I'll plot the shape of pmf function for both of the distributions for different Ns.



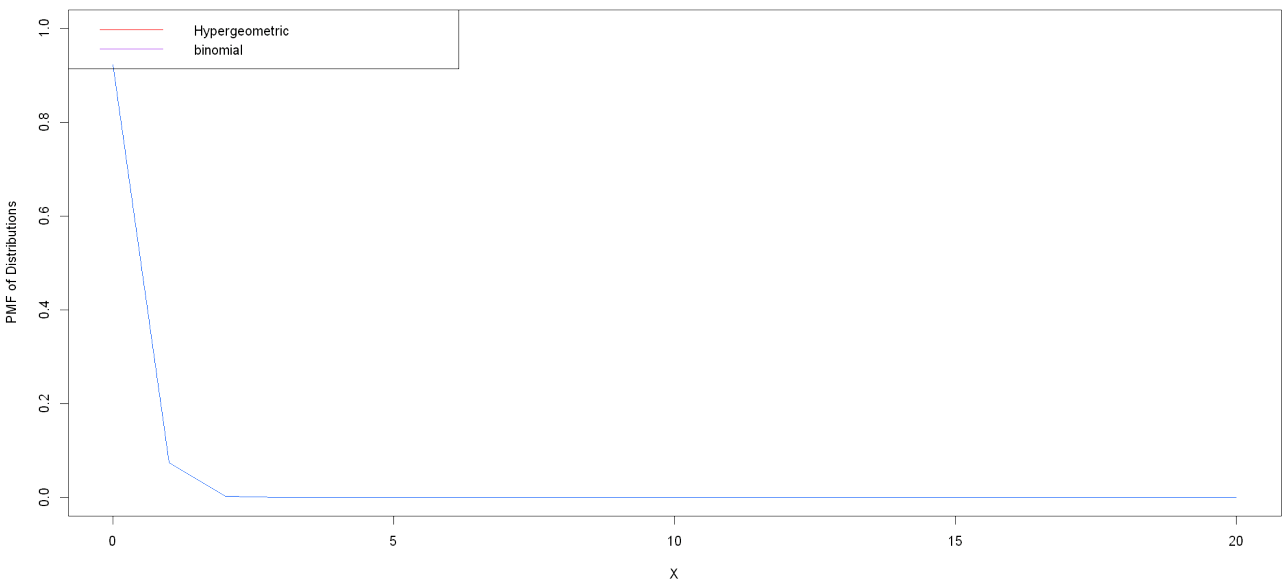
PMF Comparison for N = 500



PMF Comparison for N = 1000



PMF Comparison for N = 10000



Part 7: Analyzing Part 6

As we can see, the more we enhance the value of N the more hypergeometric distribution gets close to binomial distribution. This means when N>>n we can estimate the hypergeometric distribution with binomial distribution with n and p of K/N.

NOTE: There has to be a ration between the value of K(in my code it is num of cheaters) and N.Because if we approach N to infinity and keep K constant, the limit of p will be equal to 0 and binomial distribution won't have any meaning.

When N was equal to 100 the difference between hypergeometric and binomial distributions graphs is highly recognizable.

For N = 500 the difference decreases a lot and for N = 1000 the difference is almost not recognizable.

And finally for N = 10000 the two graphs completely overlap and you can hardly find an interval which they have differenc in.

Question 2: Binomial and Normal Distributions

Part 1: Calculating CDF For Binomial Distribution

I'll calculate the CDF by summing all the pmf values for all x less than or equal to the desired x.

```
The calculated accurate binomial distribution cdf for 430: 0.1074638
```

Part 2: Normal Estimation For Binomial Distribution

Now I estimate Bin(n, p) by N(np, np.(1-p)). To do this in the exams we have to convert P(X <= 430) to P((X - mean(X)) / std(X) <= (430 - np) / sqrt(np.(1 - p))) to be able to use the phi function(cdf for standard normal distribution). But here we have access to pnorm() function which takes mean and std as its arguments and makes it easier :).

This will have some error which I will report.

```
The estimation for cdf using normal distribution cdf: 0.1018139
The estimation error: 0.005649884
```

Part 3: Estimating With Continuity Correction

We know that if we estimate the binomial distribution which is a discrete distribution using the normal distribution which is a continuous one, then we'll have some error. According to continuity correction we can reduce the error remarkably using changing the intervals. We should only change inequalities like x <= y to x < y + 0.5. Here x <= 430 so we change it to x < 430.5.

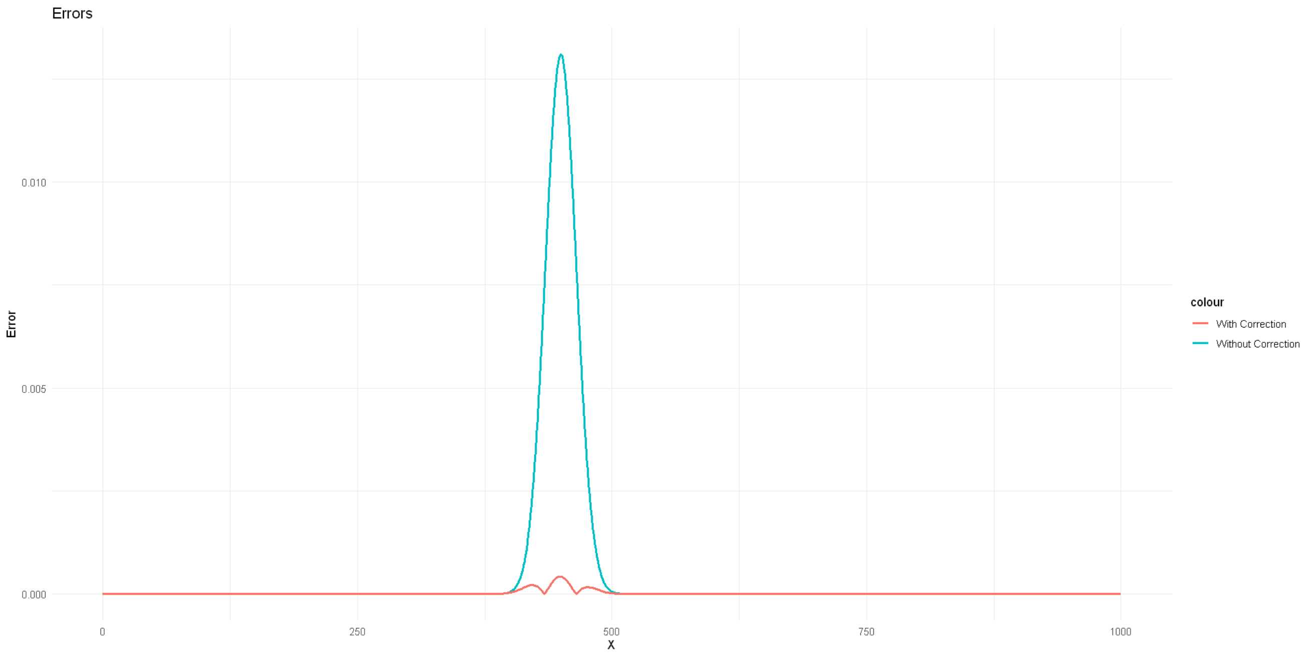
```
The estimation for cdf using normal distribution cdf: 0.1018139
The estimation error: 0.0001161443
As we can see, the corrected error is as 48.64537 times smaller than the first one.
```

Part 4: Plotting The Error Graphs

First I calculate the values for cdf by the three ways, then I calculate the error for estimations and plot them.

From now on instead of using the built-in functions of R for plotting I'll use the ggplot library which is much more professional and efficient.

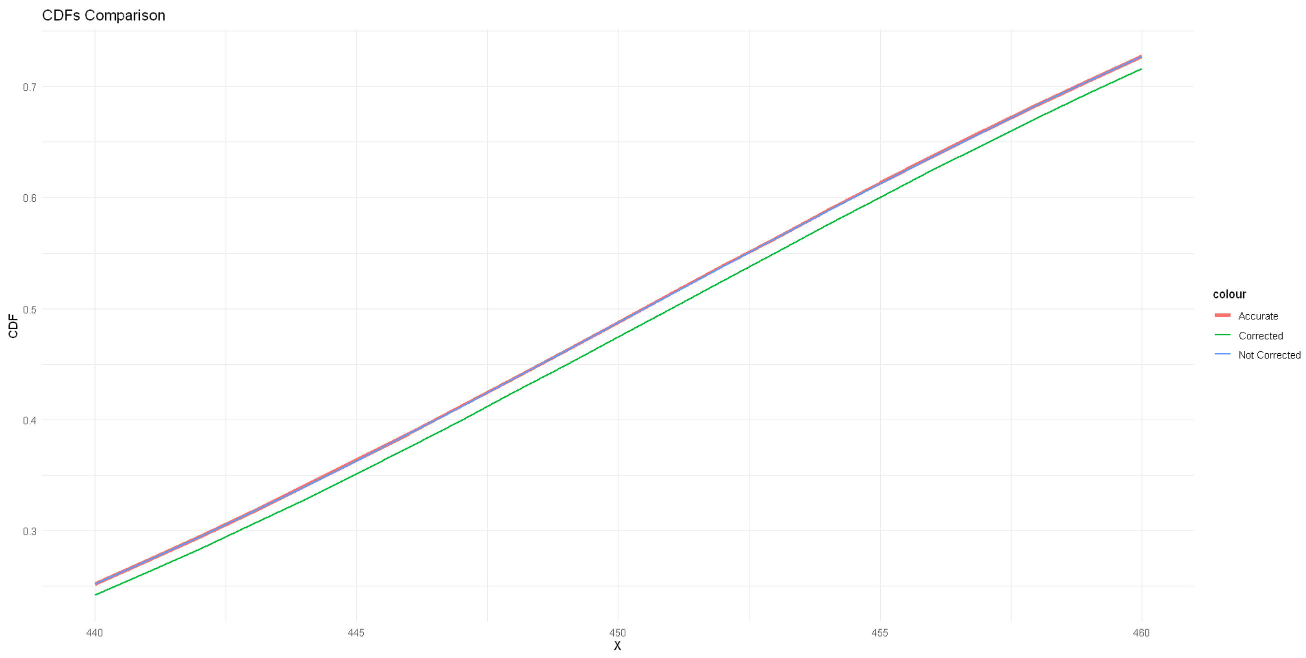
```
Warning message:
"package 'ggplot2' was built under R version 4.4.2"
Warning message:
"Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead."
```



Analyzing the two graphs proves that estimating with continuity correction has much less error than without. Also we can see that this differece has a peak about np=450 which is the expectation of the distribution. Both normal and binomial distributions have low acceleration on the low or high values of X; so in the middle points of X which is the mean, they have the most difference. This is why we have a peak up there arount 450.

Part 5: Plotting Different Versions Of CDF

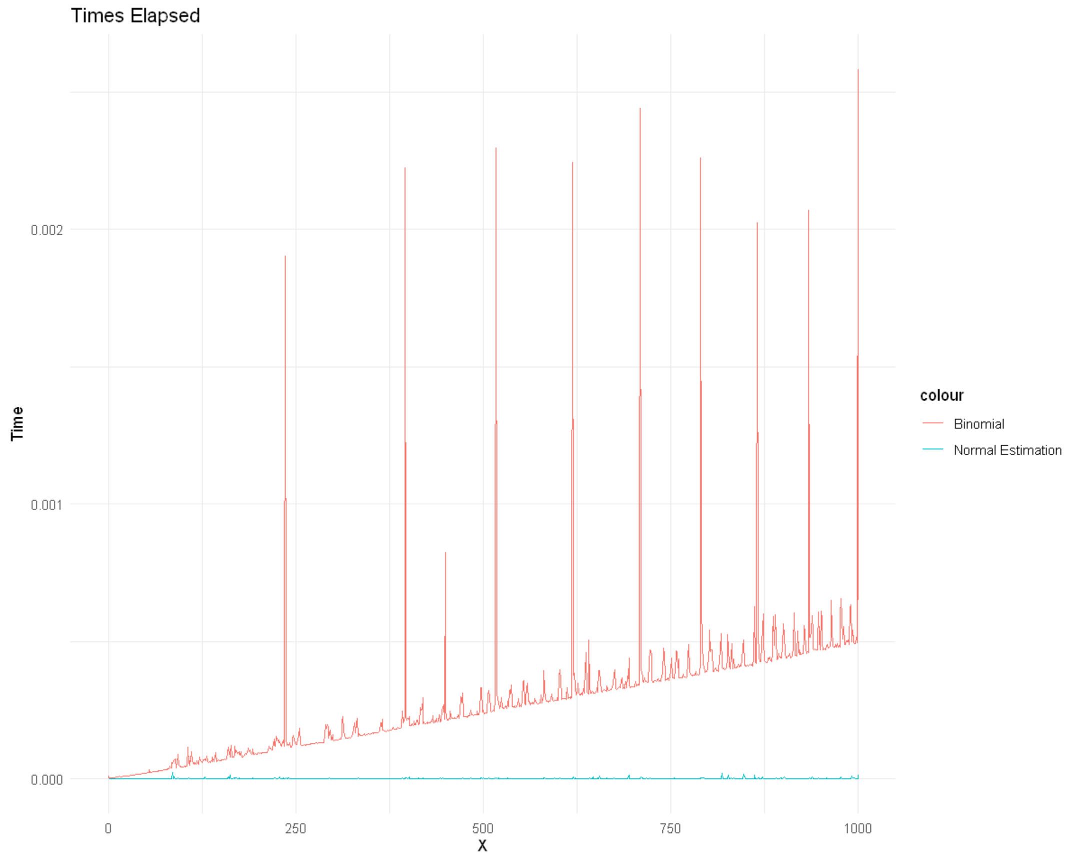
I'll plot the accurate, with and without continuity correction versions of cdf in [440, 460] interval for X.



As you can see, the corrected version is approximately identical to the accurate version. They've completely overlapped. But the not corrected version has a considerable error.

Part 6: Time Analysis For Normal and Binomial Distributions

We'll measure and compare the times that each of the binomial and normal distributions take to calculate the CDF.



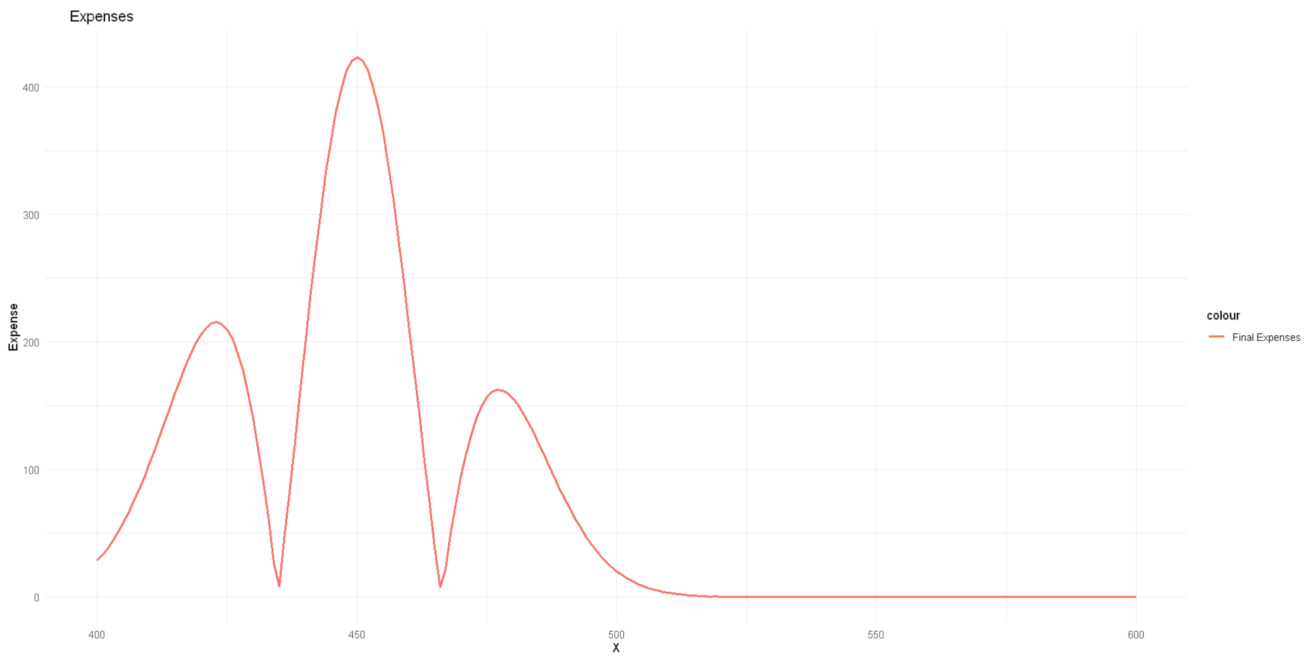
This shows that normal estimation needs much less time to calculate the cdf and the time which binomial cdf calculater needs enhances when X goes more.

Part 7: Choosing Better Method

Now we calculate the expenses and say which method is appropriate for each value of X.

529

Part 8: Calculating The Final Expenses



This graph shows that for X > 529 the expense difference between methods gets much reduced and too close to 0(it's negative.)

So the answer to the question is yes. For X = 430 calculating the CDF using the binomial method is much more rational. Becuase a large amount of error will be gained if we use the normal estimation.

Question 3: Memorylessness Of Exponential Distribution

Part 1: Generating n Samples Exponentially Distributed

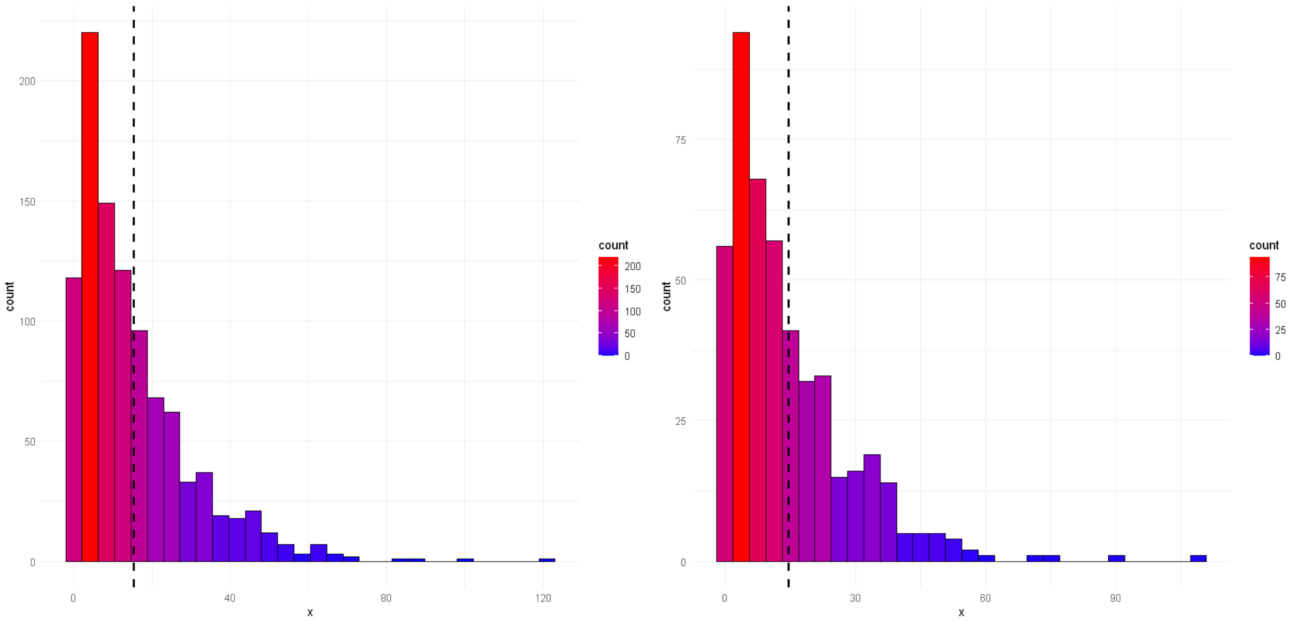
Part 2: Saving More Than 12 Minutes Time Periods

I'll iterate over the exponential samples and store each value which is between 12 and 8h * 60 = 480 minutes in another list.

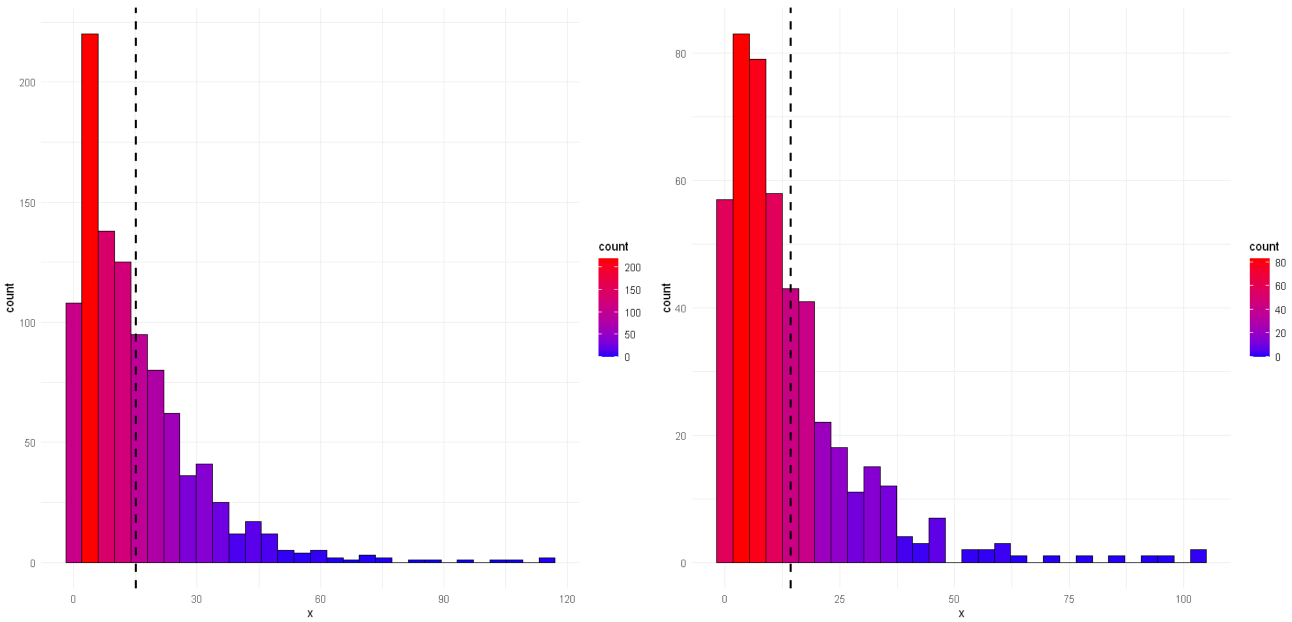
Part 3: Showing The Histograms

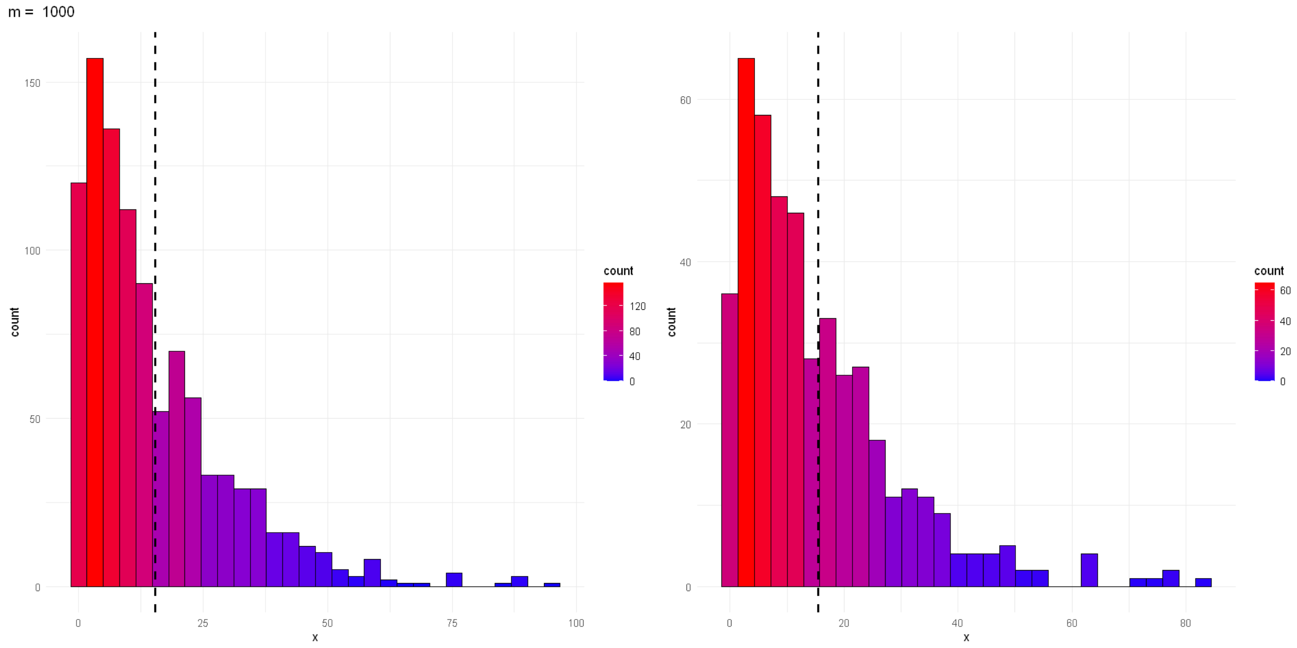
I'll plot the distribution histogram for each value of m.

m = 10



m = 100





Part 4: Analyzing The Histograms

As we can see in the histograms, the distribution histogram for times more than 12 minutes is approximately the same as the previous distribution.

Also it's obvious that when we increase m the diagrams get more identical to the standard exponential distribution diagram.

Also we can see that the mean in both graphs is almost the same in all shapes.

Part 5: Intuition Of Memorylessness

According to the memorylessness for exponential distribution $P(X > t + s \mid X > t) = P(X > s)$. So one time we simulate and another time we calculate the probability using the theoretical formula.

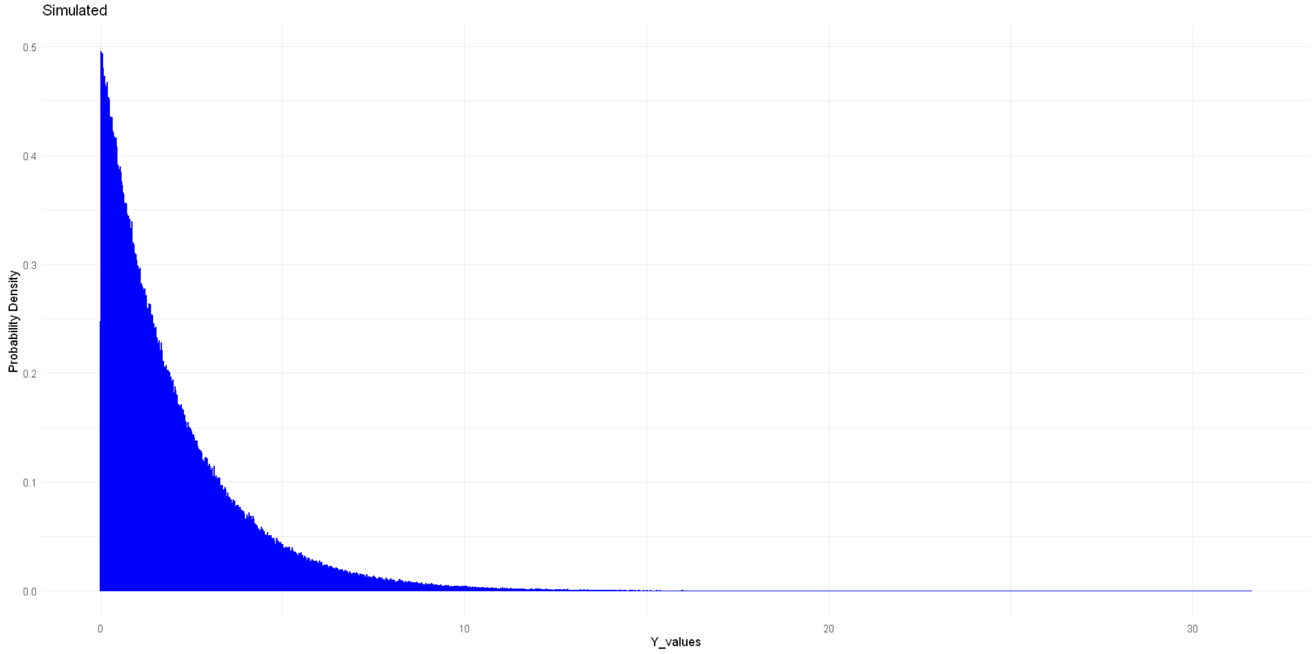
The calculated values are approximately equal.

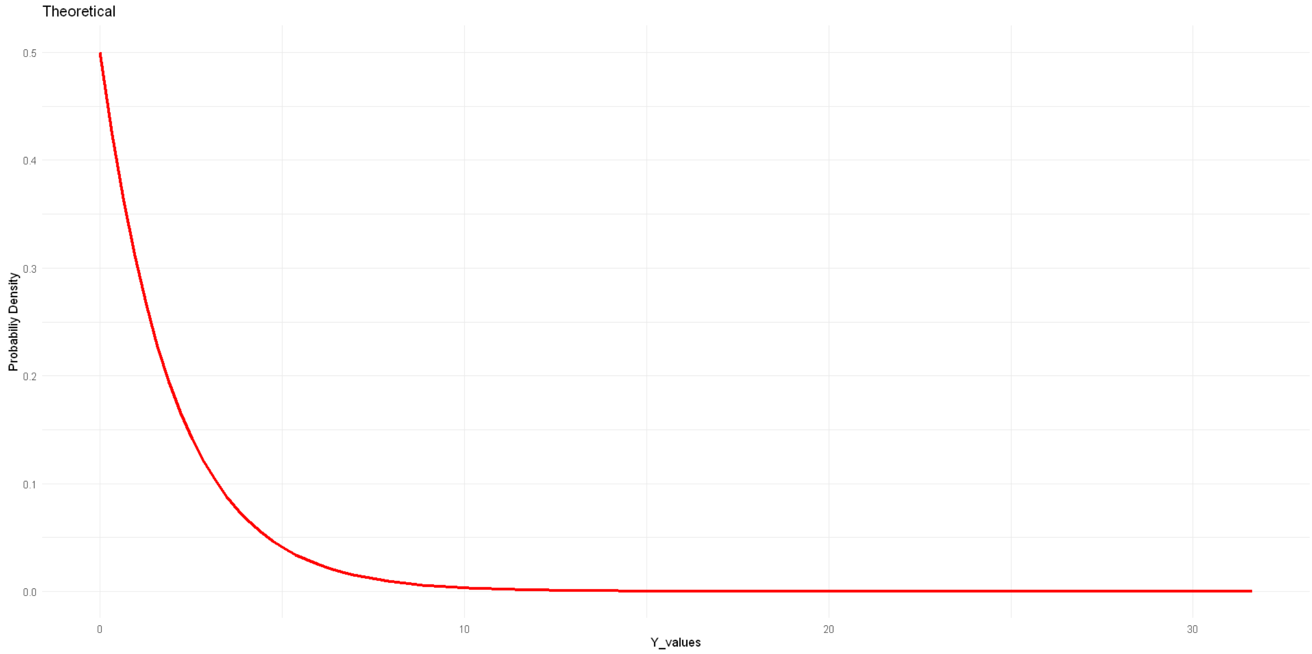
0.837259100642398
0.818730753077982

Question 4: Functions Of Random Variables

Part 1, 2 and 3: Comparing The Simulated And Theoretical Distribution For Y

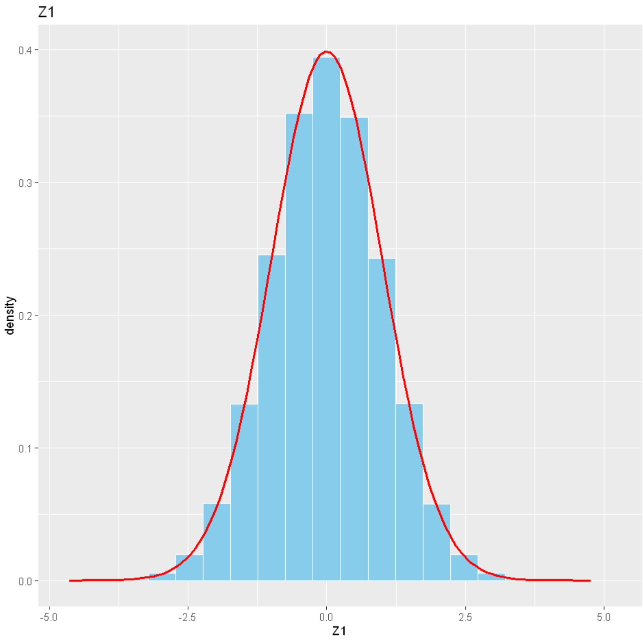
As we can see, the diagrams are approximately the same. So we get the intuition about this conversion.

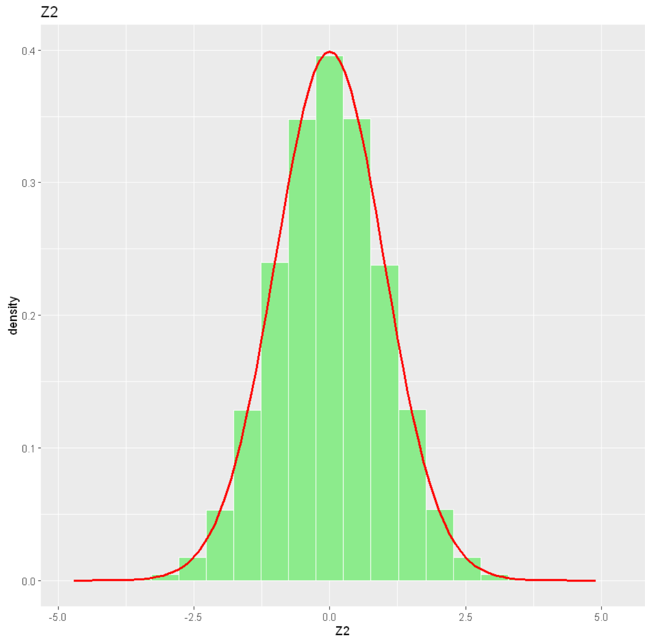




Part 4, 5 and 6: Comparing The Z1 And Z2 To Normal Distribution

As we can see again, the probability distributions for Z1 and Z2 are almost the same as standard normal distribution. This intuitively shows that the converstion works properly.


Z1
density
0.4
0.3
0.2
0.1
0.0
-5.0
-2.5
0.0
2.5
5.0
Z1


Z2
density
0.4
0.3
0.2
0.1
0.0
-5.0
-2.5
0.0
2.5
5.0
Z2